



Mögliche Fragen

Kernfusion-Reaktortechnik (Technische Universität München)



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Sample Exam Questions (MCQ) — Fusion (Lectures 1-12)

Lecture 1 — Introduction: Fusion as an Energy Source

1. The energy released in a fusion reaction primarily comes from...

- A ☐ chemical bond energy
- B ☐ the mass defect converted to energy ($E = mc^2$)
- C ☐ electron excitation
- D ☐ Coulomb repulsion energy

2. Fusion reactions release net energy mainly for nuclei...

- A ☐ heavier than iron/nickel
- B ☐ at the binding-energy peak (Fe/Ni region)
- C ☐ lighter than iron/nickel
- D ☐ only for radioactive nuclei

3. The Coulomb barrier is best described as...

- A ☐ the magnetic confinement threshold
- B ☐ electrostatic repulsion between positively charged nuclei
- C ☐ the minimum neutron energy for breeding
- D ☐ the onset of Debye shielding

4. Thermonuclear fusion means fusion driven by...

- A ☐ neutron capture
 - B ☐ gravitational compression only
 - C ☐ high particle temperature (thermal motion)
 - D ☐ electromagnetic radiation only
-

Lecture 2 — Obtaining Nuclear Fusion

5. Fusion can occur at energies far below the Coulomb barrier mainly due to...

- A ☐ relativistic time dilation
- B ☐ quantum tunneling
- C ☐ chemical catalysis
- D ☐ electron screening in metals only

6. The fusion cross section σ is a measure of...

- A ☐ fuel density
- B ☐ confinement time
- C ☐ reaction probability
- D ☐ neutron shielding quality

7. The “triple product” is...

- A ☐ $n \cdot T \cdot \tau_E$
- B ☐ $n \cdot B \cdot R$
- C ☐ $T \cdot q \cdot \beta$
- D ☐ $P_{fusion} \cdot Q \cdot \eta$

8. Ignition in a fusion plasma means...

- A ☐ external heating dominates losses
- B ☐ alpha heating balances losses (self-sustained)
- C ☐ the plasma current reaches zero
- D ☐ the vacuum vessel reaches 10^{-3} mbar

9. The fusion gain factor Q is defined as...

- A ☐ $P_{\text{external}}/P_{\text{fusion}}$
 - B ☐ $P_{\text{fusion}}/P_{\text{external}}$
 - C ☐ $P_{\text{fusion}}/P_{\text{thermal}}$
 - D ☐ $P_{\alpha}/P_{\text{neutron}}$
-

Lecture 3 — Outline of Nuclear Fusion Devices

10. A simple toroidal magnetic field alone is insufficient because...

- A ☐ it heats the plasma too much
- B ☐ it causes particle drifts leading to wall losses
- C ☐ it eliminates Debye shielding
- D ☐ it prevents any plasma current

11. The Larmor radius depends mainly on...

- A ☐ particle mass, perpendicular velocity, and magnetic field strength
- B ☐ vacuum pressure only
- C ☐ neutron energy only
- D ☐ blanket thickness only

12. In a tokamak, the poloidal field is mainly generated by...

- A ☐ only external coils
- B ☐ the plasma current
- C ☐ the divertor plates
- D ☐ cryopumps

13. A stellarator can operate steady-state primarily because...

- A ☐ it needs no vacuum vessel
- B ☐ it relies on external coils (no required plasma current)
- C ☐ it has no neutrons
- D ☐ it works at room temperature

14. One key function of the blanket is...

- A ☐ to generate the toroidal field
 - B ☐ to breed tritium and shield neutrons
 - C ☐ to pump out helium with cryopumps
 - D ☐ to induce plasma current like a transformer
-

Lecture 4 — Basics of Plasma Physics

15. A plasma is distinguished from a normal ionized gas when...

- A ☐ chemical reactions dominate
- B ☐ collective electromagnetic effects dominate
- C ☐ only neutrals are present
- D ☐ pressure is below 10^{-9} mbar

16. Debye shielding implies that...

- A ☐ the plasma is always non-neutral
- B ☐ electric fields are screened over a characteristic length λ_D
- C ☐ ions cannot move
- D ☐ magnetic fields are expelled (Meissner effect)

17. The plasma beta (β) is the ratio of...

- A ☐ magnetic pressure / plasma pressure
- B ☐ plasma pressure / magnetic pressure
- C ☐ electron pressure / ion pressure
- D ☐ confinement time / collision time

18. High-Z impurities are especially harmful because they cause strong...

- A ☐ alpha heating
 - B ☐ Bremsstrahlung radiation losses
 - C ☐ tritium breeding
 - D ☐ superconductivity
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Lecture 5 — Tokamaks: Plasma Main Operation

19. The H-mode is characterized by...

- A ☐ reduced confinement compared to L-mode
- B ☐ an edge transport barrier (pedestal)
- C ☐ no plasma current
- D ☐ operation only below 1 keV

20. A tokamak disruption typically includes...

- A ☐ only a slow density increase
 - B ☐ thermal quench followed by current quench
 - C ☐ Meissner expulsion
 - D ☐ only benign oscillations
-

Lecture 6 — Superconducting Magnets

21. The Meissner effect refers to...

- A ☐ expulsion of magnetic field from a superconductor
- B ☐ heating by plasma current
- C ☐ neutron multiplication in beryllium
- D ☐ RF resonance absorption

22. Exceeding a critical limit (T_c , B_c , or J_c) in a superconducting magnet leads to...

- A ☐ better confinement
 - B ☐ quench (normal conducting transition)
 - C ☐ higher tritium breeding
 - D ☐ lower radiation losses
-

Lecture 7 — In-Vessel Components

23. The divertor's primary purpose is to...

- A ☐ create plasma current
- B ☐ remove heat and exhaust helium ash/impurities
- C ☐ breed tritium directly in the plasma
- D ☐ reduce the Coulomb barrier

24. ITER divertor armor is typically...

- A ☐ beryllium
 - B ☐ tungsten
 - C ☐ copper
 - D ☐ 316L steel
-

Lecture 8 — Vacuum Vessel & Cryostat

25. A typical vacuum vessel pressure for fusion operation is about...

- A ☐ 1 bar
- B ☐ 10^{-3} mbar
- C ☐ 10^{-9} mbar
- D ☐ 10^3 mbar

26. The cryostat mainly provides...

- A ☐ tritium extraction
 - B ☐ thermal insulation to maintain cryogenic temperatures
 - C ☐ RF heating
 - D ☐ neutron moderation
-

Lecture 9 — Fuel Cycle & Breeding Blankets

27. The key reason tritium must be bred in the reactor is...

- A ☐ tritium is abundant in nature
- B ☐ natural supply is insufficient for reactor operation
- C ☐ tritium cannot be handled safely
- D ☐ deuterium cannot be used otherwise

28. A neutron multiplier material used in blankets is...

- A ☐ lead or beryllium
 - B ☐ helium
 - C ☐ stainless steel only
 - D ☐ graphite only
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Lecture 10 — Plasma Heating Systems

29. Ohmic heating becomes insufficient at high temperature mainly because...

- A ☐ resistivity increases with temperature
- B ☐ resistivity decreases with temperature
- C ☐ magnetic field disappears
- D ☐ Debye length becomes infinite

30. In NBI, beams are neutralized because...

- A ☐ neutrals can better follow magnetic field lines
 - B ☐ charged particles would be deflected by the magnetic field
 - C ☐ neutrals avoid all collisions
 - D ☐ neutralization increases Coulomb repulsion
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Lecture 11 — Cryogenic Technology

31. Helium is essential for many fusion magnets because...

- A ☐ it breeds tritium
- B ☐ it is the only practical coolant near 4–5 K
- C ☐ it increases plasma density
- D ☐ it improves RF coupling

32. Helium cannot be liquefied directly from room temperature via JT expansion because...

- A ☐ helium is radioactive
 - B ☐ its JT coefficient is negative at room temperature (it warms on expansion)
 - C ☐ it has too high emissivity
 - D ☐ it becomes superconducting
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Lecture 12 — Assembly & Commissioning

33. Commissioning mainly aims to...

- A ☐ build the mechanical structure only
- B ☐ verify functional performance before first plasma
- C ☐ start full-power fusion immediately
- D ☐ remove the vacuum vessel

34. Integrated testing is critical because...

- A ☐ subsystems are strongly coupled
- B ☐ plasma does not need vacuum
- C ☐ magnets can be powered warm
- D ☐ alignment does not matter